LieRing Beam Physics Manual

symplectic ring modelling with Lie maps (v0.4)

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1 LieRing Beam Physics Manual

LieRing is a modular Julia program for symplectic modelling of a storage ring (here: the Nuclotron-based lattice `magnetic_2.opt`, deuteron, 270 MeV) built on Hamiltonians and Lie algebra. Every map in the code is derived from the element Hamiltonians written in this manual; no external code was used for the physics. OptiM and COSY INFINITY results are used only as external numeric checks.

This manual describes **only what exists in the code** (status: v0.4). Each section names the source file that implements it (see Code map).

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All scripts are run from `Julia/ring` as `julia --project=. scripts/NN_name.jl`; the numbers quoted below are copied from their printed output.

2 Coordinates and reference particle

Implemented in `src/Constants.jl`.

The reference particle has charge Ze, mass m, momentum p_0, velocity $v_0 = \beta_0 c$ and Lorentz factor γ_0. The independent variable is the arc length s along the reference orbit.

User coordinates (input/output, plots):

$$u = (x,\ x',\ y,\ y',\ \phi,\ \delta), \qquad x' = \frac{dx}{ds}, \quad \delta = \frac{p - p_0}{p_0}.$$

δ is the relative *momentum* deviation (not energy). ϕ is the RF phase of arrival relative to the reference particle with the sign convention

$$\phi = -h\,\omega_0\,\Delta t, \qquad \Delta t = t - t_0(s),$$

so $\phi > 0$ means the particle arrives *early*. h is the harmonic number and $\omega_0 = 2\pi v_0/C$ the angular revolution frequency.

Canonical coordinates (all maps, Hamiltonians, symplecticity):

$$z = (x,\ p_x,\ y,\ p_y,\ l,\ \delta), \qquad p_{x,y} = \frac{P_{x,y}}{p_0}, \quad l = -v(\delta)\,\Delta t.$$

(l, δ) is a canonical pair: l is the path-length-like lag measured with the particle's own velocity $v(\delta)$, with

$$\frac{v(\delta)}{v_0} = \frac{1+\delta}{\sqrt{1 + \beta_0^2\,((1+\delta)^2 - 1)}}.$$

The energy term of every Hamiltonian is

$$e(\delta) = \frac{1}{\beta_0}\sqrt{(1+\delta)^2 + \frac{1}{\beta_0^2\gamma_0^2}} - \frac{1}{\beta_0^2},$$

(the constant is irrelevant for the dynamics and makes $e(0)$-terms vanish after removing the constant part). Particle data: deuteron, $Z = 1$, $mc^2 = 2 \times 938.272$ MeV (as in OptiM).

3 Coordinate transformation user <-> canonical

Implemented in `src/Constants.jl` *(*`canonical_to_user`*,* `user_to_canonical`*), used in* `src/TaylorMap.jl` *(*`user_map`*) and* `src/Acceleration.jl`*; checked by* `scripts/06_coordinates.jl`*.*

3.1 Exact formulas

Canonical → user:

$$p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}, \qquad x' = \frac{p_x}{p_s}, \quad y' = \frac{p_y}{p_s}, \qquad \phi = \frac{2\pi h}{C}\,\frac{l}{v(\delta)/v_0}.$$

User → canonical:

$$p_x = \frac{x'(1+\delta)}{\sqrt{1+x'^2+y'^2}}, \quad p_y = \frac{y'(1+\delta)}{\sqrt{1+x'^2+y'^2}}, \qquad l = \frac{C}{2\pi h}\,\frac{v(\delta)}{v_0}\,\phi.$$

x, y and δ are common to both sets. These follow from $x' = P_x/P_s$ and from $\phi = -h\omega_0\Delta t$, $l = -v\Delta t$.

3.2 Applied as full polynomials, not linearised

Both functions are generic Julia code. In `user_map` they are evaluated on the `TaylorN` variables, which produces the complete Taylor expansion of the square roots and of $1/v(\delta)$; the result is truncated only at the map order (`MAP_ORDER = 3`). The map in user coordinates is

$$U = T \circ M \circ T^{-1}, \qquad T = \text{canonical_to_user}.$$

Tracking (`accelerate`, `turn_snapshots`) uses the same functions on floating-point numbers, i.e. exactly. The code was checked for hidden linearisations: none were found. (The linear-optics module reports $D' = dx/d\delta$-slope in p_x; to first order $p_x = x'$, so this is not an approximation at the order it is used.)

Examples printed by `06_coordinates.jl` (order 3):

$$x' = p_x - p_x\delta + \tfrac{1}{2}p_x^3 + \tfrac{1}{2}p_x p_y^2 + p_x\delta^2 + \ldots, \qquad \phi = 0.267624\,l - 0.204533\,l\delta + 0.228642\,l\delta^2 + \ldots$$

3.3 Symplecticity holds only in canonical variables

$(x, x', y, y', \phi, \delta)$ is **not** a canonical set (x' depends on δ and p_y; ϕ is scaled by $1/v(\delta)$). Therefore the symplectic condition $J^T S J = S$ is checked only for maps in z. The user-coordinate map is the same physics written in other variables.

3.4 Numeric check (scripts/06_coordinates.jl)

check	**result**
polynomial round trip $T(T^{-1}(u)) - u$, all orders kept	max coefficient $\le 1.1 \times 10^{-16}$
exact functions round trip at $u = (5\,\text{mm}, 20\,\text{mrad}, 5\,\text{mm}, 20\,\text{mrad}, 1\,\text{rad}, 10^{-2})$	0
order-3 polynomial T vs exact T at that point	8.7×10^{-7} (truncation)
linearised T vs exact T at that point	7.6×10^{-3}
$\max \lvert J^T S J - S \rvert$ canonical one-turn map at 10^{-3}	6.8×10^{-6} (order-3 truncation)
$\max \lvert J^T S J - S \rvert$ user one-turn map at 10^{-3}	0.12 (non-canonical variables)

Selected one-turn coefficients (270 MeV, RF off):

coefficient (canonical / user)	**canonical**	**user**
$(x\|p_x)$ / $(x\|x')$	4.61348258	4.61348258
$(x\|\delta)$	2.42044391e-2	2.42044391e-2
$(l\|\delta)$ / $(\phi\|\delta)$	62.6600041	16.7693216
$(l\|\delta^2)$ / $(\phi\|\delta^2)$	-71.5275409	-31.9585141
$(l\|p_x^2)$ / $(\phi\|x'^2)$	-86.2567313	-23.0843723
$(x\|p_x^2\delta)$ / $(x\|x'^2\delta)$	343.450443	343.541752

The last line shows a genuinely nonlinear difference between the systems.

4 Element Hamiltonians and Lie maps

Types: `src/Elements.jl`*; Hamiltonians:* `src/Hamiltonians.jl`*; Lie operators:* `src/LieOperators.jl`*; factor-by-factor tracking:* `src/Tracking.jl`*.*

4.1 Hamiltonians

With $p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}$, curvature $h = 1/\rho$ and normalised strengths $k_1 = B'/(B\rho)$, $k_2 = B''/(B\rho)$, the s-Hamiltonians are (no fringe fields, hard-edge, field constant in s):

element	**Hamiltonian** $H(z)$
Drift	$e(\delta) - p_s$
Quadrupole	$e(\delta) - p_s + \frac{k_1}{2}(x^2 - y^2)$
Sextupole	$e(\delta) - p_s + \frac{k_2}{6}(x^3 - 3xy^2)$
Sector bend (with gradient)	$e(\delta) - (1+hx)p_s + hx + \frac{h^2x^2}{2} + \frac{k_1}{2}(x^2 - y^2)$
Marker	0
Thin RF kick (`RFCavity`)	kick only, see RF cavity of finite length
Finite cavity (`Cavity`)	$e(\delta) - p_s + A\cos\Phi(l, \delta, s)$ (exact integrator only; as drift in the maps)

For the bend, $\partial H/\partial x|_{z=0} = -h + h = 0$: the reference arc is a solution. All Hamiltonians are expanded as `TaylorN` polynomials to `JET_ORDER = 4` in z, and the constant term is removed.

4.2 Map of an element

For an s-independent H over length L the map is the Lie transformation

$$\mathcal{M} = \exp(-L : H :), \qquad : f : g = [f, g] = \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} \right),$$

applied to the coordinate functions as the series $\exp(: f :)z = \sum_n : f :^n z/n!$, truncated at `MAP_ORDER = 3`. The map is then Dragt–Finn factorised,

$$\mathcal{M} = R \circ e^{:f_3:} \circ e^{:f_4:},$$

R the linear (symplectic) matrix and f_3, f_4 homogeneous cubic/quartic generators. Tracking (`Tracking.jl`) applies R exactly and each $e^{:f_k:}$ by integrating the Hamiltonian flow of $-f_k$ with the implicit midpoint rule, so the tracking map is symplectic to round-off even though the generators are truncated.

5 Linear optics

`src/LinearOptics.jl`, `src/OptiMParser.jl`, script `01_linear_optics.jl`.

The linear map of each element is the Hessian of its Hamiltonian: $R = \exp(L\,S\,\mathcal{H})$, $\mathcal{H}_{ij} = \partial^2 H/\partial z_i \partial z_j|_0$. The one-turn matrix M is the product over the lattice. With $\cos\mu = \frac{1}{2}\mathrm{Tr}\,M_x$, the closed Twiss functions are

$$\beta = \frac{M_{12}}{\sin\mu}, \quad \alpha = \frac{M_{11} - M_{22}}{2\sin\mu},$$

and the closed dispersion solves $(I - M_4)D = M_{\cdot 6}$ (4x4 transverse part, $D = (D, D', D_y, D'_y)$). The momentum compaction is

$$\alpha_c = \frac{1}{\gamma_0^2} - \frac{(l|\delta)_{\text{closed}}}{C},$$

cross-checked by $\alpha_c = \frac{1}{C}\oint h D\,ds$. Twiss and dispersion are propagated in slices along the ring ("closed" check: after one turn they reproduce the start to 10^{-13}).

quantity	**LieRing**	**reference**	**source**
C [m]	117.388296		
$\beta_x(0)$ [m]	19.367522	19.3675	OptiM
$\beta_y(0)$ [m]	4.054470	4.05447	OptiM
$\alpha_x(0)$	-2.5191e-4	-2.51909e-4	OptiM
$D'(0)$	-5.0616e-4	-5.0616e-4	OptiM
$D(0)$ [m]	0.011094	0	OptiM (see Assumptions)
Q_x	2.461720	frac 0.446284	COSY
Q_y	2.461337	frac 0.476682	COSY
α_c	0.23047028	0.23036260	COSY

The tunes differ from COSY by 0.015 in both planes, with opposite signs. The reason is open (fringe-field or element-model treatment in the COSY deck is the main suspect); the OptiM initial Twiss agree to 5 digits.

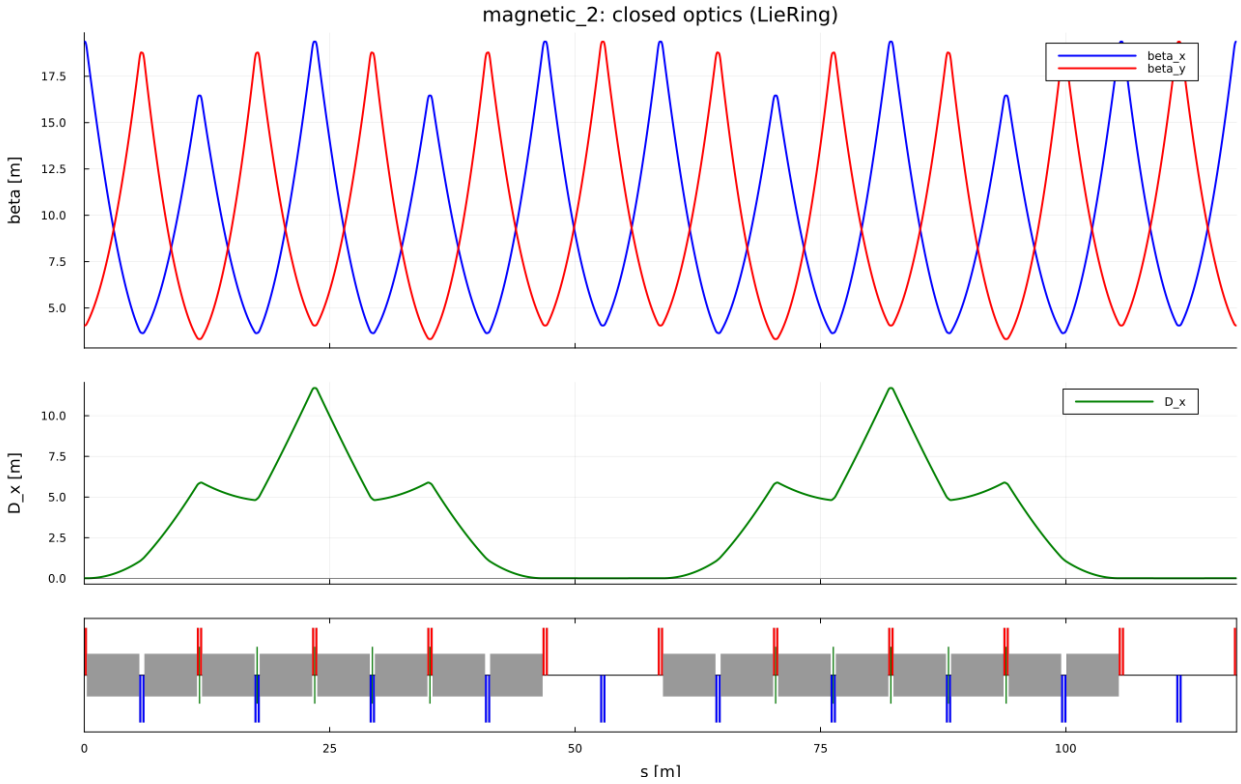


Figure 1: Closed beta functions, dispersion and magnet layout of magnetic_2 at 270 MeV (01_linear_optics.jl). Panels: beta_x and beta_y, horizontal dispersion D_x, and the magnet layout along s.

6 Second-order dispersion, momentum compaction and slip factor

`src/SecondOrder.jl`, `src/TaylorMap.jl`, script `03_second_order.jl`.

The one-turn Taylor map M (order 3, RF off) gives the closed orbit as a polynomial in δ:

$$z_\perp(\delta) = D_1\delta + D_2\delta^2 + O(\delta^3), \qquad D_1 = (D, D'),\ D_2 = (D_2, D_2'),$$

from the fixed-point condition $M_\perp(z_\perp(\delta), \delta) = z_\perp(\delta)$ order by order: $(I-R)D_1 = R_{\cdot 6}$, $(I-R)D_2 = Q(D_1\delta + \delta e_6)|_{\delta^2}$. D_2 is propagated around the ring by pushing the polynomial orbit through each element map (`dispersion_along`).

The path-length change per turn on the closed orbit is $\Delta l(\delta) = l_1\delta + l_2\delta^2$ (l component of the map evaluated on $z_\perp(\delta)$). With the velocity expansion $\frac{v_0}{v(\delta)} = 1 - c_1\delta - c_2\delta^2 + \dots$ ($c_1 = 1/\gamma_0^2$):

$$\alpha_c(\delta) = \alpha_0 + \alpha_1\delta, \quad \alpha_0 = -\frac{l_1}{C} + c_1, \quad \alpha_1 = -\frac{l_2}{C} + c_2,$$

$$\eta(\delta) = \eta_0 + \eta_1\delta, \quad \eta_0 = \alpha_0 - \frac{1}{\gamma_0^2}, \quad \eta_1 = \alpha_1 - \alpha_0 c_1 - c_2 + c_1^2.$$

η is the relative change of the revolution time, $\Delta T/T = \eta_0\delta + \eta_1\delta^2$, and enters the phase equation $\Delta\phi/\text{turn} = -2\pi h(\eta_0\delta + \eta_1\delta^2)$.

Results (natural lattice, sextupoles 0), from `05_rf_acceleration.jl` section 4 and `03_second_order.jl`:

quantity	**10 MeV**	**270 MeV**
α_0	0.23047028	0.23047028
α_1	0.33900771	0.33900771
η_0	-0.75895642	-0.53378406
η_1	1.10563172	1.01720896

α_0, α_1 are purely geometric (magnet strengths scale with p_0); the energy dependence of η comes only from c_1, c_2.

Second-order dispersion at $s = 0$ (`03_second_order.jl`): $D = 0.01109398$ m, $D' = -5.0616 \times 10^{-4}$, $D_2 = -10.054932$ m, $D_2' = 0.459066$; along the ring $D_2 \in [-10.08, 14.41]$ m; closure $D_2(C) - D_2(0) = -3.7 \times 10^{-14}$. Exact-tracking cross-check (closed orbit at fixed δ, cubic fit): $x_{co} = 0.01108483\,\delta - 10.054867\,\delta^2$, $\Delta C/C = 0.23047050\,\delta + 0.339007\,\delta^2$. Here α_1 is defined by $\Delta C/C = \alpha_0\delta + \alpha_1\delta^2$ (the local derivative is $\alpha_0 + 2\alpha_1\delta$).

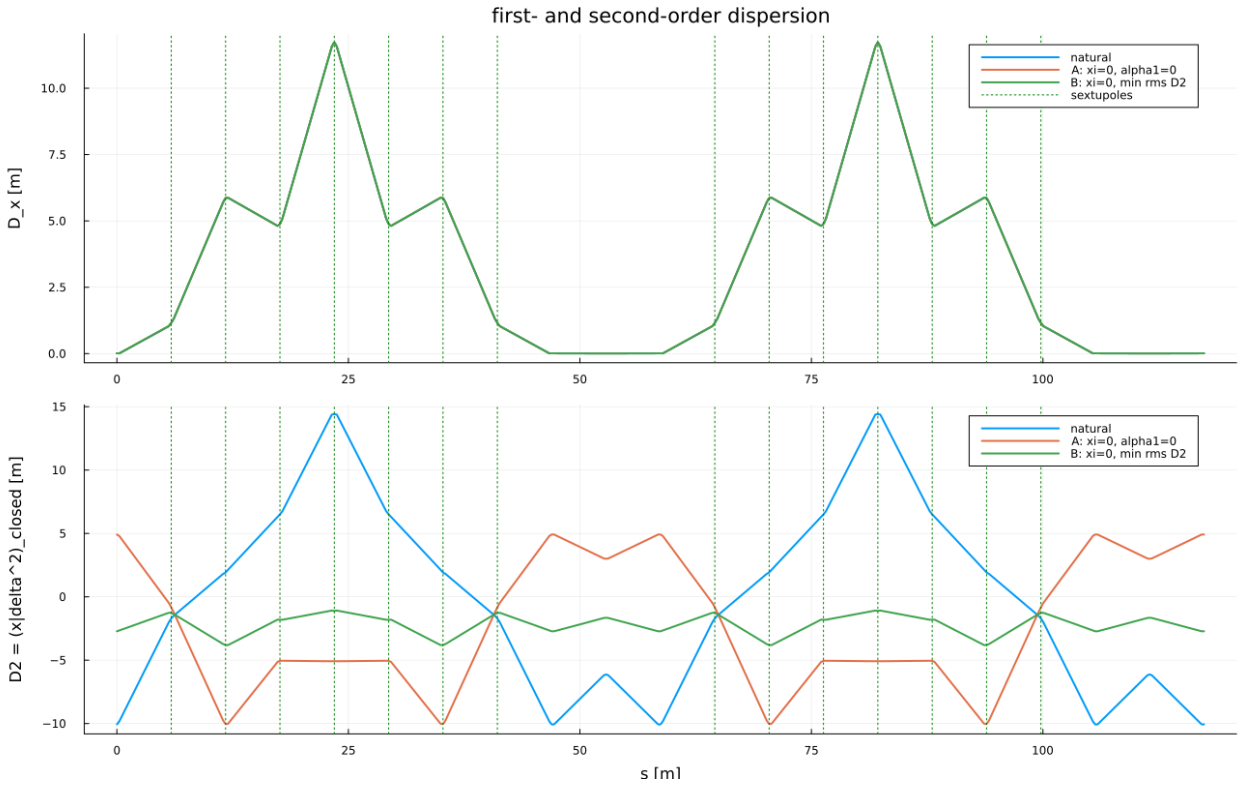


Figure 2: Second-order dispersion D2 along the ring for the natural lattice and for the optional optimiser examples (04_optional_sextupole_optimization.jl).

7 Chromaticity

`src/SecondOrder.jl` (`chromaticity_orders`), `src/TaylorMap.jl` (`tunes_at_delta`), script `03_second_order`

For fixed δ the transverse motion around the closed orbit $z_\perp(\delta)$ is linearised: the 4x4 Jacobian $J(\delta)$ of the map at the fixed point gives $Q_{x,y}(\delta)$ from $\cos 2\pi Q = \frac{1}{2}\mathrm{Tr}J$. Since δ is a constant of motion without RF, the tunes are invariants of motion and

$$Q(\delta) = Q_0 + \xi_1\delta + \tfrac{1}{2}\xi_2\delta^2, \qquad \xi_1 = \frac{dQ}{d\delta},\ \xi_2 = \frac{d^2Q}{d\delta^2},$$

obtained by central differences with $\Delta\delta = 10^{-4}$ on the polynomial map. Cross-check: tracking with the exact integrator at several δ, tune from the least-squares fit $x_{n+1} + x_{n-1} = 2\cos(2\pi Q)x_n + b$ (insensitive to a small fixed-point offset), then a parabola fit of $Q(\delta)$. Numbers: see the output of `03_second_order.jl` (reproduced in Sextupoles and the optional optimiser).

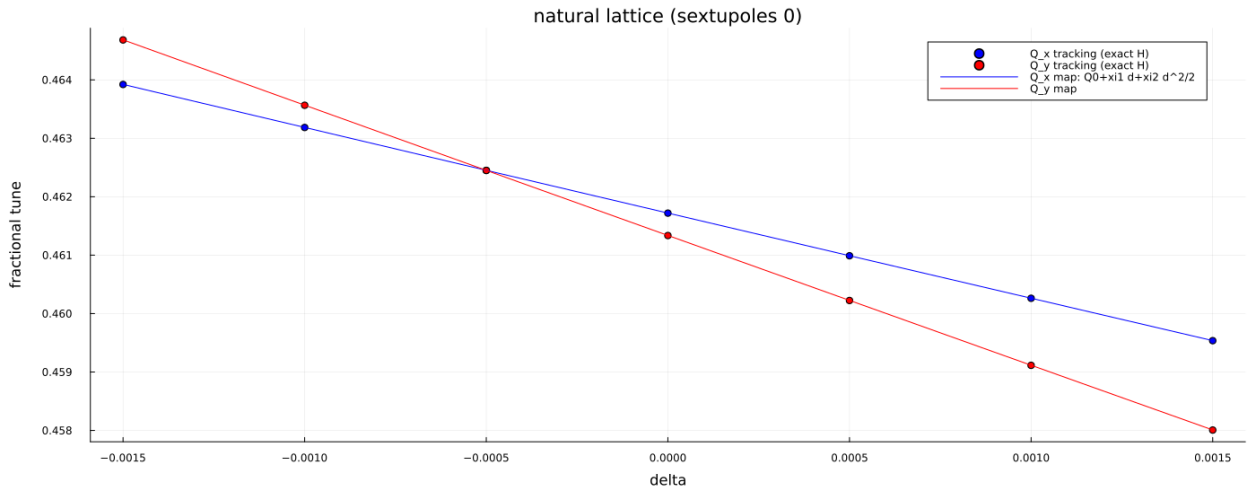


Figure 3: Tunes versus delta for the natural lattice: map (line) and tracking (markers).

	map	**tracking fit**	**COSY (reference only)**
Q_{x0} (frac)	0.46172014	0.46171903	
ξ_{1x}	-1.461538	-1.461531	-1.840801
ξ_{2x}	8.5547	8.5546	
Q_{y0} (frac)	0.46133732	0.46133566	
ξ_{1y}	-2.225488	-2.225479	-1.847679
ξ_{2y}	8.1041	8.1038	

The difference to COSY is open (its chromaticity convention was not verified).

8 RF cavity of finite length

`src/ExactIntegrator.jl` (`_field_cav`, `cavity_parameters`, `transit_factor`), type `Cavity` in `src/Elements.jl`; checks in `05_rf_acceleration.jl` section 2.

The cavity is a **finite-length** element of length L with a purely longitudinal on-axis field; it is *not* a drift plus a thin kick. With $qA_s/p_0 = -A\cos(\omega t + \psi)$, $E_s = -\partial A_s/\partial t$ and $t = s/v_0 - l/v(\delta)$, the s-Hamiltonian is

$$H_{\mathrm{cav}} = e(\delta) - p_s + A\cos\Phi, \qquad \Phi = \frac{\omega(s - L/2)}{v_0} - \frac{\omega\, l}{v(\delta)} - \phi_s,$$

$$A = \frac{ZV}{p_0 c}\,\frac{c}{\omega L}, \qquad \omega = h\omega_0 = \frac{2\pi h v_0}{C}.$$

Hamilton's equation gives

$$\frac{d\delta}{ds} = -\frac{\partial H}{\partial l} = -\frac{ZV}{p_0 c L}\,\frac{c}{v}\sin\Phi,$$

i.e. $dE/ds = -(ZV/L)\sin\Phi$. Integrated over the cavity for a particle with ϕ the energy gain is

$$\Delta E = ZVT\sin(\phi_s + \phi), \qquad T = \frac{\sin(\theta/2)}{\theta/2},\ \theta = \frac{\omega L}{v_0} = \frac{2\pi h L}{C}.$$

H_{cav} depends explicitly on s; it is integrated with the implicit midpoint rule composed to 4th order (Yoshida), which is symplectic for s-dependent Hamiltonians. Transverse RF fields are neglected.

Checks at 270 MeV ($L = 1$ m, $V = 0.2$ MV, $h = 5$, $\phi_s = 150°$, $T = 0.99702$):

check	**result**
synchronous gain vs $ZVT\sin\phi_s$	0.099700856 vs 0.099701840 MeV
thin limit $L = 0.1/0.01/0.001$ m vs thin kick ($\phi = 0.05$)	-6.3×10^{-9}, -1.5×10^{-10}, -9.9×10^{-12} in δ
symplectic error of the cavity Jacobian	4.4×10^{-16}

The thin kick (`RFCavity`, `Tracking.rf_kick`) applies the same energy gain $ZV\sin(\phi_s + \phi)$ at one point; the finite cavity converges to it as $L \to 0$ (table above).

9 Acceleration and adiabatic damping

`src/ExactIntegrator.jl` (`renormalize`), `src/Acceleration.jl` (`accelerate`, `turn_snapshots`), script `05_rf_acceleration.jl`.

9.1 Reference momentum update

All particles, including the synchronous particle (#1, initially $z = 0$), are tracked through the full lattice with the exact integrator. Right after the cavity the synchronous particle has $\delta_s \neq 0$; it is made the new reference:

$$p_0' = p_0(1+\delta_s), \quad 1+\delta' = \frac{1+\delta}{1+\delta_s}, \quad p_x' = \frac{p_x}{1+\delta_s},\ p_y' = \frac{p_y}{1+\delta_s}, \quad \Delta t' = \Delta t - \Delta t_s,\ l' = -v\,\Delta t'.$$

Positions x, y are unchanged. The magnets are assumed to ramp with p_0 (normalised h, k_1, k_2 fixed) and the RF frequency follows $h\,\omega_0(p_0)$ (locked to the synchronous particle).

9.2 Adiabatic damping

The physical transverse momentum $P_x = p_0 p_x$ is not changed by a longitudinal field, so $p_x \propto 1/p_0$. Between kicks the motion is a betatron oscillation with fixed optics; the Courant–Snyder action in normalised variables $J = \frac{1}{2}(\gamma x^2 + 2\alpha x p_x + \beta p_x^2)$ is an adiabatic invariant of the *unnormalised* phase space, $\oint P_x\,dx = p_0 \oint p_x\,dx = 2\pi p_0 J = \text{const}$. Hence $J \propto 1/p_0$ and the amplitude $A = \sqrt{2\beta J}$ scales as

$$\frac{A(n)}{A(0)} = \sqrt{\frac{p_{\mathrm{inj}}}{p_0(n)}}.$$

Ramp results (10 → 270 MeV, $V = 0.2$ MV, $h = 5$, $\phi_s = 150°$, then 1000 turns with $\phi_s = 180°$): see the table in Longitudinal motion, separatrix and synchrotron tune.

10 Longitudinal motion, separatrix and synchrotron tune

`src/SecondOrder.jl` (`LongitudinalParams`, `long_H`, `fixed_points`, `separatrix`, `bucket_height`, `synchrotron_tune`), plots in `src/Plotting.jl`, script `05_rf_acceleration.jl`.

10.1 Turn-by-turn map and smooth Hamiltonian

With one cavity per turn, the motion of (ϕ, δ) per turn n is

$$\phi_{n+1} = \phi_n - 2\pi h\left(\eta_0\delta_{n+1} + \eta_1\delta_{n+1}^2\right), \qquad \delta_{n+1} = \delta_n + A\left(\sin(\phi_s+\phi_n) - \sin\phi_s\right), \quad A = \frac{ZVT}{\beta_0^2 E_0}.$$

(The tracking itself does not use this map; it integrates the exact Hamiltonians element by element. The map and the Hamiltonian below are used for formulas and for drawing the separatrix.) Treating n as continuous,

$$H(\phi,\delta) = -2\pi h\left(\frac{\eta_0\delta^2}{2} + \frac{\eta_1\delta^3}{3}\right) + A\left(\cos(\phi_s+\phi) + \phi\sin\phi_s\right).$$

Fixed points on $\delta = 0$: stable $\phi = 0$, unstable $\phi_u = \pi - 2\phi_s$; η_1 adds a second pair at $\delta \approx -\eta_0/\eta_1$ (far outside the bucket here, 0.69 at 10 MeV). The **separatrix** is the level line $H(\phi,\delta) = H(\phi_u, 0)$; the bucket is the region around $(0,0)$ inside it. Stability requires $\eta_0\cos\phi_s > 0$: below transition ($\eta_0 < 0$) and with our sign of ϕ this means $90^\circ < \phi_s \le 180^\circ$; acceleration uses $\phi_s = 150^\circ$ (gain $ZVT\sin\phi_s > 0$), the flat top $\phi_s = 180^\circ$.

10.2 Synchrotron tune

Small amplitude:

$$Q_s^{\text{smooth}} = \sqrt{\frac{h\,\eta_0\,ZVT\cos\phi_s}{2\pi\beta_0^2 E_0}}, \qquad \cos(2\pi Q_s^{\text{kick}}) = 1 - \frac{(2\pi Q_s^{\text{smooth}})^2}{2} \quad \text{(one kick per turn)}.$$

E **[MeV]**, $\phi_s = 180^\circ$	**tracking**	**smooth**	**kick**
10	0.0784653	0.0777018	0.0784950
270	0.0129407	0.0129376	0.0129411

At 10 MeV Q_s is large and the discrete-kick formula is needed (1% vs 0.04%).

10.3 Ramp 10 -> 270 MeV

Parameters: $L = 1$ m, $V = 0.2$ MV, $h = 5$ (f_{RF} 1.313 -> 6.200 MHz), $\phi_s = 150^\circ$; gain 0.099702 MeV/turn; 270 MeV reached at turn 2608, then 1000 turns at $\phi_s = 180^\circ$. Bucket half-height at injection $+0.02425/-0.02369$ (asymmetric because of η_1), stationary bucket at 270 MeV 0.00976.

result	**value**
tracked	64 particles x 3608 turns (330 s)
mean gain per turn	0.099698 MeV (expected 0.099702)
x amplitude 2 mm ->	0.8701 mm (ratio 0.43506)
y amplitude 2 mm ->	0.8658 mm (ratio 0.43291)
prediction $\sqrt{p_{inj}/p_{fin}}$	0.43142
ensemble particles lost from the bucket	0 of 60
longitudinal invariant (turn 300 -> 2493)	1.0000 -> 0.9508

The first ~200 turns are not adiabatic (gain = 1% of E_{kin} per turn at 10 MeV), which explains most of the 5% drift of the longitudinal invariant.

Ensemble: 60 particles with $\phi \sim N(0, 0.6)$ rad, $\delta \sim N(0, 0.25\,\hat{\delta}_{\text{bucket}})$, kept if inside the injection separatrix, on their dispersion orbit, $x' = y' = 0$. This is deliberately mismatched to the bucket so that filamentation is visible. η_1 and α_1 enter the separatrix; $\alpha_0 = 0.23047028$, $\alpha_1 = 0.33900771$ from the ring map (energy independent, as expected for scaled magnets).

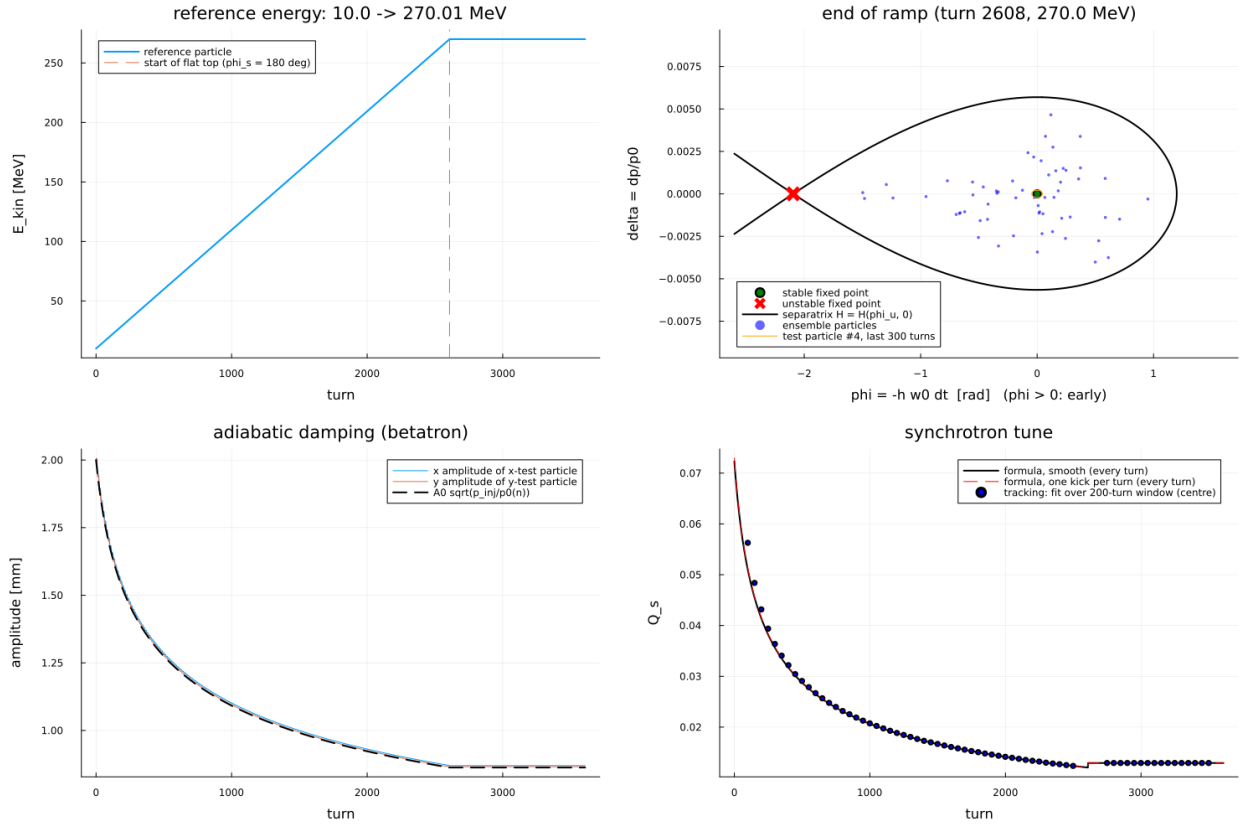


Figure 4: Acceleration overview (05_rf_acceleration.jl). Top left: reference kinetic energy vs turn; dashed line: start of the flat top. Top right: longitudinal phase space at the end of the ramp (turn 2608); black: analytic separatrix H = H(phi_u,0) of the smooth Hamiltonian with eta0, eta1 at that energy and phi_s = 150 deg; red cross: unstable fixed point; green: stable fixed point; blue dots: the 60 ensemble particles; orange: test particle #4 (delta0 = 1e-3) over the last 300 turns (small, near the centre). Bottom left: betatron amplitudes of the x and y test particles vs turn and the prediction A0 sqrt(p_inj/p0(n)). Bottom right: synchrotron tune; lines: smooth and one-kick formulas evaluated at every turn (step at turn 2608 where phi_s changes to 180 deg); markers: tracking, tune fitted to delta of particle #4 in 200-turn windows moved by 50 turns, plotted at the window centre (windows containing the ramp end are skipped).

Animations (HTML only; the PDF shows the static figure): `assets/longitudinal_ramp.gif` - ensemble every 20 turns during the ramp and flat top, with the separatrix of the current energy and ϕ_s; right panel with δ divided by the current bucket half-height (removes the adiabatic shrinking and shows the filamentation). `assets/longitudinal_elements.gif` - one turn at injection, frame = exit of each element; right panel: change since the turn start (only the cavity changes δ; ϕ changes in every element through η-like path/velocity terms).

Animation assets/longitudinal_ramp.gif (see the HTML manual or the GIF file): Animation: ensemble during the ramp (GIF)

Animation assets/longitudinal_elements.gif (see the HTML manual or the GIF file):

Animation: one turn element by element (GIF)

11 Symplecticity checks and Taylor-map comparison

`src/Symplectic.jl`, `src/TaylorMap.jl`, `src/LieOperators.jl`, script `02_lie_maps.jl`.

A map $z \mapsto \mathcal{M}(z)$ is symplectic iff its Jacobian satisfies $J^T S J = S$ with $S = \mathrm{diag}(S_2, S_2, S_2)$, $S_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The code reports $\max |J^T S J - S|$ for

- every element Lie map (at $z = 0$ and at a finite amplitude),
- the ring map built from Lie factors, the factor-by-factor tracking map

(Jacobian computed exactly by running the tracker on `TaylorN`), and the exact integrator (including the cavity).

Truncated polynomial maps are symplectic only up to the truncation order (the error grows like $|z|^{\mathrm{MAP_ORDER}}$); the Lie-factor tracker and the exact integrator are symplectic to round-off.

Taylor-map comparison module. `TaylorMap.jl` builds the one-turn map completely independently of the Lie machinery: Hamilton's equations $dz/ds = S\nabla H$ are integrated with RK4 in the truncated polynomial algebra (a COSY-style differential-algebra map), element by element, and the maps are composed. Its coefficients are compared with the Lie-factor product and with one-turn tracking. Numbers: printed by `02_lie_maps.jl` (see Sextupoles and the optional optimiser for the tables reproduced from the latest run).

Latest run of `02_lie_maps.jl` / `03_second_order.jl`:

check	**value**
linear ring matrix: Lie product vs product of $\exp(LSA)$	2.3×10^{-13}
linear ring matrix: Taylor/RK4 vs $\exp(LSA)$	6.9×10^{-9}
max 2nd-order coefficient difference Lie vs Taylor (largest coeff 167.5)	5.1×10^{-8}
symplectic error ring Taylor map at 0 / at finite amplitude	6.8×10^{-13} / 1.4×10^{-5}
one turn: Lie tracking vs Taylor map	4.7×10^{-10}
symplectic error of 1-turn Lie-tracking Jacobian	2.4×10^{-14}
one turn: exact integrator vs Lie (order 3) / vs Taylor / vs linear	2.4×10^{-8} / 2.4×10^{-8} / 6.9×10^{-5}
symplectic error of exact-integrator one-turn Jacobian	1.1×10^{-13}
1000 turns Lie tracking: relative spread of J_x, J_y	2.4×10^{-3}, 2.7×10^{-3}

12 Sextupoles and the optional optimiser

Placement (lattice editing, no physics): `src/Elements.jl` *(*`find_optics_maxima`*,* `place_sextupole_families`*,* `set_family_strengths`*). Optimisation:* `src/Optimizers.jl`*, called* ***only*** *by* `scripts/04_optional_sextupole`

Default: all sextupoles have the strength given in the lattice file, which is zero for `magnetic_2.opt`. No script except 04 changes them.

12.1 Placement

Three families of 0.12 m sextupoles are placed into the nearest existing sextupole slot or short drift next to local maxima (≥ 0.8 of the global maximum) of D_x (family SXD, 2 magnets), β_x (SXB, 4) and β_y (SYB, 8). A sextupole acts chromatically through $x = D\delta$: the first-order chromaticity contribution is $\Delta\xi_{x,y} = \pm\frac{1}{4\pi}\sum k_2 L\,\beta_{x,y} D$, so SXD/SXB act mostly on ξ_x and SYB on ξ_y; D_2 and α_1 are driven by $k_2 L D^2$. The circumference is unchanged.

12.2 Response (per unit $k_2 = 1$ $\mathbf{m}^{-3}$ for the whole family)

family	$\Delta\xi_{1x}$	$\Delta\xi_{1y}$	$\Delta\alpha_1$	$\Delta D_2(0)$ **[m]**
SXD	4.32833	-0.90618	-1.63792	89.86
SXB	3.67887	-0.74301	-0.40970	-0.034
SYB	0.84330	-4.33701	-0.24240	9.455

$\Delta\eta_1 = \Delta\alpha_1$ (sextupoles do not change c_1, c_2).

12.3 Optional example solutions

These targets are *examples*, not requirements: **A** $\xi_{1x} = \xi_{1y} = \alpha_1 = 0$; **B** $\xi_{1x} = \xi_{1y} = 0$ and minimum rms D_2 (least squares).

quantity	**natural**	**A**	**B**
k_2 SXD / SXB / SYB [m^{-3}]	0 / 0 / 0	0.2304 / 0.2653 / -0.6067	0.1456 / 0.3650 / -0.6061
ξ_{1x}	-1.4615	0.0000	0.0000
ξ_{1y}	-2.2255	0.0000	0.0000
ξ_{2x}	8.555	3.934	5.165
ξ_{2y}	8.104	-0.717	1.236
$D_2(0)$ [m]	-10.055	4.905	-2.718
max $\lvert D_2\rvert$ [m]	14.41	10.04	3.81
rms D_2 [m]	7.558	5.728	2.411
α_1	0.33901	0.00000	0.09799
η_1 (270 MeV)	1.01721	0.67820	0.77619

Tracking check of A: $\xi_{1x} = 0.00001$, $\xi_{2x} = 3.934$, $\xi_{1y} = 0.00000$, $\xi_{2y} = -0.717$ (map: identical to 3 digits).

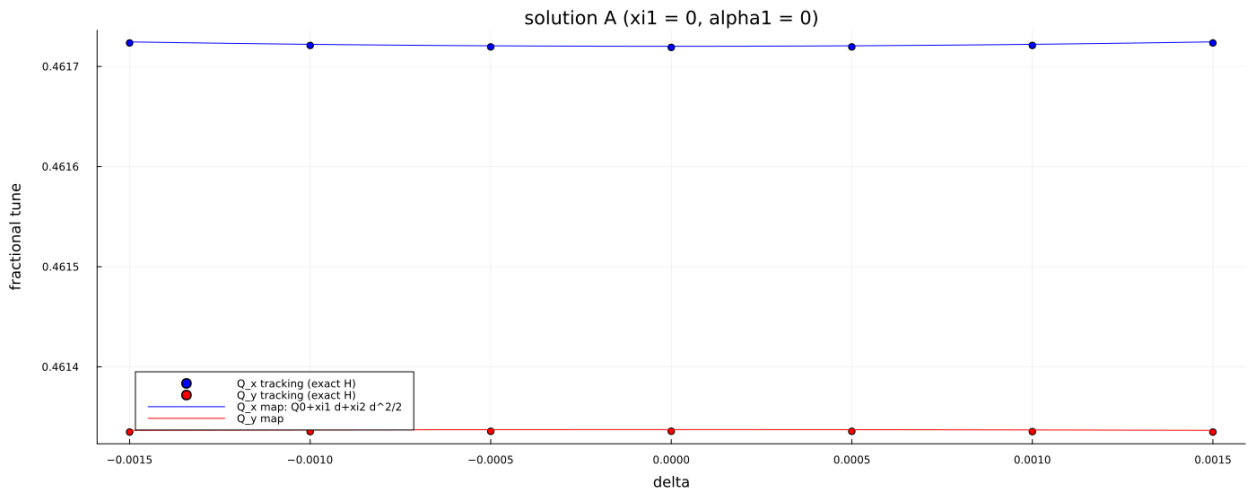


Figure 5: Tunes vs delta after the optional correction A: map (lines) and exact tracking (markers).

13 Assumptions and hypotheses

Every choice that is not a consequence of the lattice file or of the equations is listed here.

1. **Injection energy 10 MeV** (5 MeV/u): not in `magnetic_2.opt`; chosen as a typical linac energy and confirmed by the user. Final 270 MeV (135 MeV/u) is from the file header.
2. **RF parameters** (chosen, not from a design): cavity length $L = 1$ m, voltage $V = 0.2$ MV, harmonic $h = 5$, $\phi_s = 150°$ during the ramp and 180° on the flat top, one cavity in the first `OBM` drift. V is large so that the ramp takes ~2600 turns instead of 10^5-10^6.
3. **Cavity field** purely longitudinal on axis, sinusoidal in time, uniform over L; **no transverse RF fields**, no beam loading.
4. **Magnets ramp with p_0** (normalised strengths fixed); RF frequency locked to $h\omega_0(p_0)$ of the synchronous particle.
5. **No fringe fields**, hard-edge elements, sector bends; **no space charge**, no impedances, no radiation, no errors/misalignments.
6. **Test amplitudes**: $x_0 = 2$ mm, $y_0 = 2$ mm, $\delta_0 = 10^{-3}$ ($x' = y' = 0$); ensemble $\phi \sim N(0, 0.6)$, $\delta \sim N(0, 0.25\hat{\delta})$, 60 particles, seed 1.
7. **Optimisation targets A and B** are optional examples; the default lattice has sextupoles at zero.
8. **"Number of periods = 2"** in `magnetic_2.opt`: the listed sequence already bends by 360 deg, so it is used **once** (as in the COSY deck).
9. **$D(0) = 0$ in OptiM** initial conditions vs our closed $D(0) = 0.0111$ m: OptiM's value is an input, not the closed solution; all other initial Twiss agree to 10^{-5}.
10. Mass 2×938.272 MeV (as OptiM), not the deuteron mass 1875.613 MeV.
11. α_1 used for the separatrix is linearly interpolated between the maps at 10 and 270 MeV (it turned out equal at both).
12. Map orders: Hamiltonians to order 4, maps to order 3; tracking with the exact integrator uses $ds = 0.2$ m sub-steps (4th order).

14 Code map

file	physics implemented
`src/Constants.jl`	constants, deuteron, reference particle, $v(\delta)$, $e(\delta)$, exact user <-> canonical transform
`src/Jets.jl`	truncated polynomial space (TaylorSeries `TaylorN`), orders
`src/Elements.jl`	**all** element types (Drift, SBend, Quadrupole, Sextupole, thin RFCavity, finite Cavity, Marker), Lattice, lattice-editing helpers (sextupole placement, cavity insertion)
`src/Hamiltonians.jl`	element Hamiltonians as polynomials
`src/LieOperators.jl`	Poisson bracket, exp(: f :), Dragt–Finn factors R, f_3, f_4
`src/OptiMParser.jl`	reader of OptiM `.opt` files
`src/LinearOptics.jl`	linear maps, closed Twiss, dispersion, tunes, α_c
`src/Symplectic.jl`	$J^T S J - S$ checks
`src/Tracking.jl`	factor-by-factor Lie tracking, thin RF kick
`src/TaylorMap.jl`	independent DA (RK4) one-turn Taylor map, closed orbit, tunes vs δ
`src/NormalForm.jl`	linear normalisation (nonlinear normal form: planned)
`src/SecondOrder.jl`	D_2, $\alpha_{0,1}$, $\eta_{0,1}$, $\xi_{1,2}$, longitudinal Hamiltonian, separatrix, Q_s
`src/ExactIntegrator.jl`	exact-H symplectic integrator, finite RF cavity, reference renormalisation
`src/Acceleration.jl`	ramp loop, tune fit, element-by-element snapshots
`src/Plotting.jl`	all figures and GIFs
`src/Optimizers.jl`	OPTIONAL sextupole response and example solutions
`scripts/01_linear_optics.jl`	linear optics + OptiM/COSY comparison
`scripts/02_lie_maps.jl`	Lie maps, symplecticity, Lie vs Taylor, tracking
`scripts/03_second_order.jl`	D_2, α/η, chromaticities
`scripts/04_optional_sextupole_optimization.jl`	optional optimiser
`scripts/05_rf_acceleration.jl`	RF cavity, ramp, separatrix, GIFs
`scripts/06_coordinates.jl`	coordinate transformation checks

15 Programmer manual (stub)

To be written. Entry point: `src/LieRing.jl` includes all modules in dependency order and re-exports their names; scripts do `include("../src/LieRing.jl"); using .LieRing`. Build these docs with `julia --project=docs docs/make.jl` (HTML in `docs/build/`) and the PDF with `julia docs/make_pdf.jl` (needs xelatex).